Phantom: Distributional Forecasting for Cross-Sectional Cryptocurrency Return Prediction

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Abstract

Cross-sectional return prediction—ranking which assets will outperform their peers—is a cornerstone of quantitative investing, yet remains underexplored in cryptocurrency markets where distributional properties (heavy tails, regime shifts) demand more than point forecasts. We present Phantom, a Transformer-based model that predicts full Student-t distributions of cross-sectional *relative* returns across 362 cryptocurrencies at horizons of 1–30 days, using only standard OHLCV price and volume features. The model employs a patched encoder with cross-attention decoding for simultaneous multi-horizon prediction, trained on cross-sectionally demeaned returns to isolate idiosyncratic signal from shared market factors. On 15 months of fully out-of-sample data (January 2025–March 2026), Phantom achieves a rank information coefficient (IC) of 0.124 at the 10-day horizon (t = 8.93, Newey-West), with a long-short portfolio Sharpe ratio of 11.84 net of 30 basis points transaction costs, a 74.7% daily win rate, and perfect decile monotonicity. Double-sorted portfolio analysis reveals that Phantom's signal is concentrated in high-volatility assets (IC = 0.092, t = 6.77) and is partially correlated with the low-volatility anomaly (ρ = 0.63), yet combining both signals yields the highest IC of 0.157. Through systematic ablation across eight model versions, we establish that the cross-sectional signal resides in daily OHLCV patterns, cannot be extracted from absolute returns, does not benefit from derivative market features (funding rates, order flow), and scales with the breadth of the asset universe—but not with higher-frequency data.

Keywords

cryptocurrency, cross-sectional prediction, distributional forecasting, Transformer, Student-t

1 Introduction

The cryptocurrency market, with a combined capitalization exceeding $2 trillion and continuous 24/7 trading across hundreds of liquid assets, presents a distinctive opportunity for cross-sectional return prediction. Unlike directional forecasting—predicting whether an asset's price will rise or fall—cross-sectional prediction concerns the simpler but economically powerful task of *ranking*: identifying which assets will outperform or underperform their peers over a given horizon. Recent work has established that cross-sectional factors familiar from equity markets, including momentum [1], size, and volume, carry significant premia in cryptocurrency [2]. However, existing approaches rely exclusively on point forecasts, discarding the distributional information that is critical for position sizing, risk management, and tail-risk awareness in a market characterized by heavy tails, extreme volatility clustering, and frequent regime shifts.

This paper addresses a gap at the intersection of distributional forecasting and cross-sectional prediction. While probabilistic time series models such as DeepAR and Temporal Fusion Transformers produce predictive distributions for individual assets, they are not designed for cross-sectional ranking. Conversely, cross-sectional factor models [1, 2] produce rankings but provide no uncertainty quantification. Distributional foundation models such as JointFM [5] have shown that Transformer architectures can recover complex distributional structure from time series, but have only been evaluated on synthetic data with no demonstrated cross-sectional signal on real financial assets. We bridge these approaches.

We present **Phantom**, a Transformer-based model that predicts calibrated Student-t distributions of cross-sectional *relative* returns across 362 cryptocurrencies at horizons of 1–30 days simultaneously. The architecture consists of a patched Transformer encoder that processes 120 days of 6-channel OHLCV features, a cross-attention decoder that queries all 30 horizons in a single forward pass, and a Student-t output head with 3 parameters per horizon (location μ, scale σ, degrees of freedom ν). The model is trained on cross-sectionally demeaned returns—each asset's return minus the same-day cross-sectional mean—which removes shared market factors and isolates the idiosyncratic signal that drives relative performance.

On 15 months of fully out-of-sample data (January 2025 to March 2026, 430 trading days), Phantom achieves a rank information coefficient (IC) of 0.124 at the 10-day horizon (Newey-West t = 8.93), increasing to 0.166 at 30 days. A long-short portfolio that buys the top quintile and shorts the bottom quintile earns a Sharpe ratio of 11.84 after 30 basis points of transaction costs per trade, with a 74.7% daily win rate, perfect decile return monotonicity (Spearman ρ = 1.0), and IC positive in all 15 out-of-sample months. The strategy's breakeven transaction cost is 336 basis points, providing substantial headroom above typical exchange fees.

These results emerge from a systematic search over the design space, spanning eight model versions that progressively identify where cross-sectional signal resides—and where it does not.

Contributions.

- We propose Phantom, the first distributional forecasting model for cross-sectional cryptocurrency return prediction, producing calibrated Student-t distributions at 30 horizons simultaneously.
- Through systematic ablation across eight model versions, we establish that (i) cross-sectional signal exists in daily OHLCV features but only in *relative* (not absolute) returns, consistent with weak-form market efficiency; (ii) synthetic SDE pretraining achieves oracle accuracy on synthetic data but

transfers zero signal to real assets; (iii) derivative market features (funding rates, taker buy ratios) provide no incremental IC; and (iv) higher-frequency (4-hour) data degrades performance due to sample overlap, establishing that the signal is daily-scale.
- We demonstrate that the signal scales with asset universe breadth (Sharpe doubles from 60 to 362 assets) and survives transaction costs up to 336 basis points per trade.
- We characterize the signal's relationship to the low-volatility anomaly via double-sorted portfolios: Phantom's predictive power concentrates in high-volatility assets (IC = 0.092, $t = 6.77$) where low-vol sorting alone provides no ranking, and combining both signals achieves the highest IC of 0.157.

2 Related Work

2.1 Cross-Sectional Cryptocurrency Factors

Liu, Tsyvinski, and Wu [1] establish a three-factor model for cryptocurrency returns comprising market (CMKT), size (CSMB), and momentum (CMOM) factors, constructed analogously to the Fama-French framework for equities. Momentum emerges as the dominant cross-sectional predictor, with weekly long-short returns and t-statistics well above 3, substantially exceeding the momentum premium observed in equity markets. Their work demonstrates that standard asset pricing methodology transfers to the crypto asset class, though the factor zoo is far smaller than in equities. Fieberg et al. [2] extend this line of work by constructing CTREND, a composite trend factor that aggregates momentum and moving-average crossover signals across multiple lookback horizons. CTREND subsumes any individual momentum measure and earns significant alpha over the three-factor model, demonstrating that multi-horizon trend aggregation is more informative than any single-horizon signal.

The low-volatility anomaly (low-volatility assets earn higher risk-adjusted returns than high-volatility ones) is well-documented in equities [9] and has been observed in cryptocurrency markets, where the retail-dominated investor base and lottery-like preferences for high-volatility tokens provide a natural explanation. Our double-sort analysis (Section 5.4) reveals substantial interaction between our model's signal and the low-volatility effect.

All existing cross-sectional crypto studies produce point forecasts: they rank assets by a scalar signal and evaluate via portfolio sorts. None provides distributional predictions that quantify the *uncertainty* of the ranking, which is essential for position sizing and risk management.

2.2 Transformer-Based Time Series Forecasting

Nie et al. [3] introduce PatchTST, which segments time series into subseries-level patches before applying a Transformer encoder, reducing sequence length while capturing local semantic information. Their finding that channel-independent processing outperforms channel mixing on standard benchmarks provides a foundation for our patched encoder design, though we depart from channel independence by jointly embedding all six OHLCV features within each patch, as cross-channel interactions (e.g., between returns and volume) are central to our prediction task. Ansari et al. [4] propose Chronos, a time series foundation model that tokenizes real-valued observations into discrete bins and trains a T5-based language model on a large corpus of diverse time series, enabling zero-shot probabilistic forecasting across domains. While both PatchTST and Chronos advance time series forecasting, neither is designed for cross-sectional ranking or the heavy-tailed distributional properties specific to financial returns.

2.3 Distributional Foundation Models

Most closely related to our work, Hackmann [5] proposes JointFM, a foundation model for multivariate distributional time series forecasting. JointFM uses Variable-Factored Attention to model both temporal and cross-variable dependencies and employs parametric distributional heads—multivariate Gaussian mixtures and multivariate Skewed Student-t distributions—to produce full joint predictive distributions over multiple coupled targets. Trained on synthetic data generated from stochastic differential equations (GBM, Merton jump-diffusion, Bates stochastic volatility, regime-switching), JointFM achieves strong distributional calibration as measured by energy distance [8]. However, JointFM remains evaluated *exclusively* on synthetic processes and does not address real financial data, leaving open the question of whether synthetic pretraining transfers to actual asset returns. Our work answers this question directly: through systematic comparison of synthetic and real-data pretraining (Section 5.7), we find that synthetic SDE pretraining achieves oracle distributional accuracy on synthetic test data but transfers zero cross-sectional signal to real cryptocurrency markets. Phantom extends the distributional forecasting paradigm to real cross-sectional prediction with demonstrated out-of-sample signal.

2.4 Probabilistic Forecasting

Distributional forecasting of financial returns has a long history, from GARCH-family models to deep learning approaches including DeepAR and Temporal Fusion Transformers. We evaluate distributional quality using the Continuous Ranked Probability Score (CRPS), a strictly proper scoring rule [6]:

$$\mathrm{CRPS}(F, y) = \mathbb{E}_F[|X - y|] - \tfrac{1}{2}\,\mathbb{E}_F[|X - X'|] \tag{1}$$

where X, X' are independent draws from the predictive distribution F and y is the realized value. Strict propriety ensures that the score is uniquely minimized when the forecaster reports the true conditional distribution, providing a principled training and evaluation objective.

We parameterize predictive distributions as Student-$t(\mu, \sigma, \nu)$ rather than Gaussian for two reasons: (i) cryptocurrency returns exhibit substantial excess kurtosis even at multi-day horizons, and (ii) the degrees-of-freedom parameter ν provides an adaptive, learnable measure of tail heaviness per asset and horizon. Our finding that the model consistently pushes $\nu \to 2$ (the minimum for finite variance) confirms that heavy-tailed modeling is essential for this asset class and that Gaussian assumptions are inadequate.

3 Methodology

3.1 Problem Formulation

Let $\mathcal{A}_t = \{1, \ldots, N_t\}$ denote the set of actively traded cryptocurrency assets on date t. For each asset i, we observe a context window of daily OHLCV (open, high, low, close, volume) data spanning

$L = 120$ trading days, from which we compute a feature matrix $\mathbf{x}_i \in \mathbb{R}^{L \times C}$ with $C = 6$ channels (Section 3.2). We define the *cumulative log return* of asset i over the next h days as

$$r_{i,t,h} = \sum_{k=1}^{h} \log\left(\frac{c_{i,t+k}}{c_{i,t+k-1}}\right) \tag{2}$$

where $c_{i,t}$ is the closing price. To isolate idiosyncratic signal from shared market movements, we work with *relative returns* (Section 3.5):

$$r^{\text{rel}}_{i,t,h} = r_{i,t,h} - \frac{1}{N_t} \sum_{j \in \mathcal{A}_t} r_{j,t,h} \tag{3}$$

The model f_θ produces a parametric predictive distribution for each asset and horizon simultaneously:

$$f_\theta(\mathbf{x}_i) = \left\{(\mu_{i,h},\ \sigma_{i,h},\ \nu_{i,h})\right\}_{h=1}^{H}, \quad H = 30 \tag{4}$$

where each triplet parameterizes a Student-t distribution: $r^{\text{rel}}_{i,t,h} \sim$ Student-$t(\mu_{i,h},\ \sigma_{i,h},\ \nu_{i,h})$. The model serves a dual objective: (1) *calibrated distributional forecasts*—the predicted distributions should match the empirical frequency of outcomes—and (2) *cross-sectional ranking signal*—on each date, the predicted location parameters $\{\mu_{i,h}\}_{i \in \mathcal{A}_t}$ should rank assets in an order correlated with their actual relative performance.

3.2 Feature Engineering

From raw OHLCV data $(o_t, h_t, l_t, c_t, v_t)$ for each asset, we compute six channels that are universal across all tradeable assets:

Table 1: Input feature channels.

Ch	Feature	Formula
0	Log return	$\log(c_t/c_{t-1})$
1	Intraday range	$(h_t - l_t)/c_t$
2	Body ratio	$(c_t - o_t)/(h_t - l_t)$
3	Log volume ratio	$\log(v_t/\widetilde{v}_{30})$
4	Trailing volatility	$\hat{\sigma}_{30} \cdot \sqrt{252}$
5	Momentum	$\sum_{k=0}^{9} r_{t-k}$

Here $\widetilde{v}_{30}$ is the 30-day rolling median volume and $\hat{\sigma}_{30}$ is the 30-day rolling standard deviation of log returns, annualized by $\sqrt{252}$. Channels 1–2 capture intraday price dynamics (volatility and candle structure), channels 3–4 capture volume and realized volatility, and channel 5 captures short-term trend. Critically, we apply *no instance normalization* (e.g., RevIN [10]) or mean-absolute scaling, as our ablation studies (Section 5.7) found that such normalizations destroy the cross-sectional signal by removing the relative magnitude differences between assets that the model exploits for ranking.

3.3 Model Architecture

Patched encoder. The input $\mathbf{x}_i \in \mathbb{R}^{L \times C}$ is divided into $N = L/P$ non-overlapping patches of length $P = 5$ time steps, giving $N = 24$ patches. Each patch $\mathbf{p}_n \in \mathbb{R}^{P \times C}$ is flattened and linearly projected into $\mathbb{R}^d$:

$$\mathbf{z}_n^{(0)} = \mathbf{W}_{\text{proj}} \cdot \text{flatten}(\mathbf{p}_n) + \mathbf{e}_n \tag{5}$$

where $\mathbf{W}_{\text{proj}} \in \mathbb{R}^{d \times PC}$ is the projection matrix and $\mathbf{e}_n \in \mathbb{R}^d$ is a learned positional encoding, with $d = 512$. The resulting sequence $\mathbf{Z}^{(0)} \in \mathbb{R}^{N \times d}$ is processed by an M-layer pre-norm Transformer encoder ($M = 8$, GELU activation, $d_{\text{ff}} = 2048$, 8 attention heads), producing contextualized patch representations $\mathbf{Z}^{(M)} \in \mathbb{R}^{N \times d}$. The full architecture is shown in Figure 1.

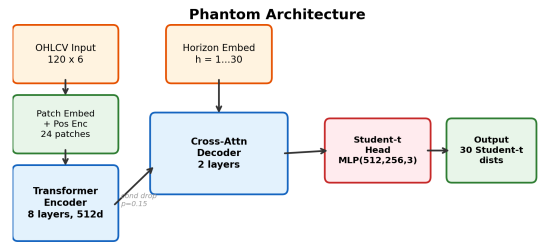

Figure 1: Phantom architecture. The patched encoder processes 120 days of 6-channel OHLCV input into 24 contextualized patch tokens. The cross-attention decoder simultaneously queries all 30 horizons against the encoder output. Each decoded vector is mapped to Student-t parameters via a two-layer MLP.

Cross-attention decoder. To decode distributions at all $H = 30$ horizons in a single forward pass, each horizon $h \in \{1, \ldots, H\}$ is mapped to a query $\mathbf{q}_h^{(0)} = \mathbf{E}_h$ via a learned embedding table. The 30 queries attend to the encoder output through $D = 2$ decoder layers, each applying pre-norm cross-attention followed by a feedforward network:

$$\hat{\mathbf{q}}_h^{(\ell)} = \mathbf{q}_h^{(\ell-1)} + \text{MultiHead}\Big(\underbrace{\text{LN}(\mathbf{q}_h^{(\ell-1)})}_{\text{query}},\ \underbrace{\mathbf{Z}^{(M)}}_{\text{key, value}}\Big) \tag{6}$$

$$\mathbf{q}_h^{(\ell)} = \hat{\mathbf{q}}_h^{(\ell)} + \text{FFN}\Big(\text{LN}(\hat{\mathbf{q}}_h^{(\ell)})\Big)$$

This produces a decoded vector $\mathbf{q}_h^{(D)} \in \mathbb{R}^d$ per horizon, with total complexity $O(H \cdot N \cdot d)$.

Student-t output head. The decoded vector $\mathbf{q}_h^{(D)} \in \mathbb{R}^d$ for each horizon is mapped to distribution parameters via a two-layer MLP:

$$\begin{aligned}(\hat{\mu},\ \log\hat{\sigma},\ \log\hat{\nu}) &= \mathbf{W}_2\,\text{GELU}(\mathbf{W}_1\,\mathbf{q}_h^{(D)} + \mathbf{b}_1) + \mathbf{b}_2 \\ \sigma_h &= \exp(\text{clamp}(\log\hat{\sigma},\ -7,\ 2)) \\ \nu_h &= \exp(\text{clamp}(\log\hat{\nu},\ \log 2.01,\ \log 100))\end{aligned} \tag{7}$$

The clamping ensures $\sigma_h \in [e^{-7}, e^2]$ and $\nu_h \in [2.01, 100]$. The lower bound $\nu > 2$ guarantees finite variance; our ablation finds the model consistently pushes $\nu \approx 2$, indicating maximally heavy-tailed distributions.

Condition dropout. During training, the encoder output is zeroed with probability $p_{\text{drop}} = 0.15$:

$$\tilde{\mathbf{Z}}^{(M)} = \mathbf{Z}^{(M)} \cdot \mathbf{m}, \quad m_i \sim \text{Bernoulli}(1 - p_{\text{drop}}) \tag{8}$$

applied per sample (not per token). This trains the decoder to produce reasonable predictions even without encoder input, preventing

the decoder from learning to ignore the encoder via its residual connection—an analog of classifier-free guidance in diffusion models.

3.4 Training Objective

The loss function combines four terms:

$$\mathcal{L} = \lambda_{\text{NLL}}\,\mathcal{L}_{\text{NLL}} + \lambda_{\text{CRPS}}\,\mathcal{L}_{\text{CRPS}} + \lambda_{\text{MSE}}\,\mathcal{L}_{\text{MSE}} + \lambda_{\text{var}}\,\mathcal{L}_{\text{var}} \tag{9}$$

with weights $\lambda_{\text{NLL}} = 1.0$, $\lambda_{\text{CRPS}} = 0.5$, $\lambda_{\text{MSE}} = 0.3$, $\lambda_{\text{var}} = 0.1$.

Negative log-likelihood. The primary loss is the NLL of the observed relative return under the predicted Student-t distribution, averaged across all horizons and samples:

$$\mathcal{L}_{\text{NLL}} = -\frac{1}{BH}\sum_{b,h} \log\, p\Big(r^{\text{rel}}_{b,h} \mid \mu_{b,h}, \sigma_{b,h}, \nu_{b,h}\Big) \tag{10}$$

where $p(\cdot \mid \mu, \sigma, \nu)$ is the Student-t density.

CRPS.. The CRPS term (Eq. 1) is evaluated on a subset of horizons $\{1, 5, 10, 15, 20, 25, 30\}$ days using the sample-based representation with 256 Monte Carlo draws from the predicted distribution.

Mean MSE with horizon weighting. To encourage non-trivial conditional means (preventing collapse to $\mu = 0$), we add a weighted MSE loss on the predicted mean:

$$\mathcal{L}_{\text{MSE}} = \frac{1}{B}\sum_{b}\sum_{h=h_{\min}}^{H} w_h \left(\mu_{b,h} - r^{\text{rel}}_{b,h}\right)^2, \quad w_h = \sqrt{h/H} \tag{11}$$

The $\sqrt{h}$ weighting reflects the signal-to-noise ratio of drift: for a random walk with drift δ, $\text{SNR}(h) = \delta\sqrt{h}/\sigma \propto \sqrt{h}$. We set $h_{\min} = 5$ to avoid penalizing mean predictions at very short horizons where SNR is negligible.

Encoder variance penalty. To prevent representation collapse (all encoder outputs becoming identical regardless of input), we add a penalty on the variance of encoder representations across the batch: $\mathcal{L}_{\text{var}} = -\text{Var}_b[\bar{\mathbf{z}}_b]$ where $\bar{\mathbf{z}}_b$ is the mean-pooled encoder output for sample b.

3.5 Cross-Sectional Demeaning

The relative return (Eq. 3) removes the shared market factor from each asset's return on each date. If all assets rise 5% but asset i rises 8%, its relative return is +3%; an asset rising only 2% receives −3%. This is analogous to the market-adjustment step in Fama-MacBeth regressions and serves two purposes: (1) it makes the prediction task *market-neutral*, so the model need not predict market direction, and (2) it isolates the *idiosyncratic* component of returns that drives cross-sectional dispersion.

The demeaning is computed offline over all (t, h) pairs before training. We require a minimum of 3 assets per date; 99.9% of samples are adjusted. The demeaned target has mean approximately zero by construction, but its *cross-sectional variance* varies substantially across dates and horizons—this variance is what the model's $\sigma_{i,h}$ learns to predict.

4 Experimental Setup

4.1 Data

We collect daily OHLCV data for all actively traded USDT-denominated pairs on Binance. After filtering stablecoins, leveraged tokens, and assets with fewer than 300 days of history, our universe contains **362 cryptocurrencies** spanning August 2017 to March 2026.

Table 2: Dataset statistics.

	Train	Validation	Test
Period	< 2024-01	2024	≥ 2025-01
Samples	229,759	96,966	144,630
Unique dates	2,208	366	430
Assets per date	∼160	∼290	∼340

The temporal split ensures strict out-of-sample evaluation: the validation year (2024) acts as a buffer between training and test data. The model is initialized from a checkpoint pretrained on a broader universe of 413 assets (including equities, forex, and commodities) before fine-tuning on the crypto-only dataset.

4.2 Baselines

We compare against the following cross-sectional strategies, all evaluated using the same long-short portfolio methodology (top/bottom quintile, equal-weighted, daily rebalancing):

- **Momentum** (**kd**): rank assets by cumulative log return over the past $k \in \{5, 10, 20\}$ days. This is the dominant factor in the crypto literature [1].
- **Mean-reversion** (**kd**): rank by *negative* past $k \in \{1, 5\}$ day return.
- **Volume**: rank by mean log volume ratio (channel 3) over the past 5 days.
- **Low volatility**: rank by *negative* trailing realized volatility (channel 4). The low-volatility anomaly is well-documented across asset classes [9].
- **Random**: Gaussian noise signal (null hypothesis).

4.3 Evaluation Metrics

Cross-sectional signal. The primary metric is the *rank information coefficient* (IC): the Spearman rank correlation between predicted $\mu_{i,h}$ and realized $r^{\text{rel}}_{i,h}$, computed per date and averaged. We report Newey-West adjusted t-statistics [7] throughout to account for autocorrelation in daily IC and return series. The *long-short portfolio* buys the top quintile (∼72 assets) and shorts the bottom quintile by predicted μ, equal-weighted, rebalanced daily. We report annualized Sharpe, Sortino, and Calmar ratios using a 365-day convention (crypto trades 24/7).

Transaction costs. We compute net returns under one-way cost assumptions of 0, 5, 10, 20, 30, and 50 basis points, applied proportionally to *turnover*—the fraction of portfolio positions that change each day. We report the *breakeven cost*: the maximum one-way fee at which the strategy's mean daily return reaches zero.

Distributional quality. We evaluate calibration via the probability integral transform (PIT): if the predicted distribution F_θ is correctly calibrated, the PIT values $F_\theta(r^{\text{rel}})$ should be uniformly distributed. We report coverage at the 50%, 80%, 90%, and 95% levels and the Kolmogorov-Smirnov test statistic for PIT uniformity.

5 Results

All results in this section are on the fully out-of-sample test set (January 2025 to March 2026, 430 trading days, 144,630 samples across 362 assets).

5.1 Cross-Sectional Signal

Figure 2(a) shows the rank IC by forecast horizon. IC increases monotonically from 0.080 at the 1-day horizon to 0.166 at 30 days, consistent with the theoretical scaling of drift signal-to-noise ratio with $\sqrt{h}$. At the reference 10-day horizon, the mean IC is 0.124 with a Newey-West t-statistic of 8.93, and IC is positive on 78% of trading days and in all 15 out-of-sample months.

5.2 Portfolio Performance

Table 3 presents the quintile portfolio sort at the 10-day horizon. Returns increase monotonically from Q1 (−1.49%/day) to Q5 (+1.04%/day), yielding a long-short spread of 2.53%/day with a Newey-West t-statistic of 7.44. The Spearman rank correlation between quintile rank and mean return is 1.000 (perfect monotonicity).

Table 3: Quintile portfolio sort (h=10d). Newey-West t-statistics in parentheses.

	Q1	Q2	Q3	Q4	Q5	Q5−Q1
Return (%/d)	−1.49	−0.24	0.18	0.48	1.04	2.53
	(−7.71)	(−1.80)	(1.83)	(4.64)	(5.05)	(7.44)
Sharpe	−13.4	−2.8	2.6	6.9	9.0	13.0

Figure 2(b) shows the cumulative long-short returns under different cost scenarios. The gross annualized Sharpe is 13.0; after 30 bps of transaction costs per trade, the net Sharpe remains 11.84 with cumulative returns of 993%. The breakeven one-way cost is 336 bps, far exceeding Binance's standard fee of 10 bps. The high Sharpe reflects both the model's IC and the diversification benefit of trading 362 assets: for a long-short strategy, Sharpe scales approximately as IC $\times \sqrt{N}$ where N is the number of assets. Average daily portfolio turnover is 37.8%.

5.3 Baseline Comparison

Table 5 compares Phantom to simple cross-sectional strategies. All momentum signals have *negative* IC during the test period, indicating a mean-reverting regime in the 2025–2026 crypto market—consistent with the literature finding that crypto momentum is time-varying [2]. Mean-reversion signals are weakly positive but not statistically significant. The strongest baseline is the **low-volatility** strategy (IC = 0.165), which we analyze further in Section 5.4.

Table 4: Strategy performance summary (h=10d, top/bottom 20%, equal-weighted).

Metric	Value	Metric	Value
Ann. return	925%	Win rate	74.7%
Ann. volatility	71%	Max drawdown	−38.6%
Sharpe (gross)	13.00	Avg. turnover	37.8%
Sharpe (10 bps)	12.61	Breakeven cost	336 bps
Sharpe (30 bps)	11.84	IC (NW t)	0.124 (8.93)
Sortino	27.3	L/S ret (NW t)	7.45
Calmar	24.0	IC > 0 days	78%

Table 5: Baseline comparison (h=10d).

Strategy	Rank IC	Sharpe	Win%
Phantom (ours)	**0.124**	**13.0**	**74.7**
Momentum (5d)	−0.007	−1.0	49.1
Momentum (10d)	−0.012	−1.0	45.8
Momentum (20d)	−0.016	−0.8	48.6
Mean-rev (1d)	0.009	0.8	50.0
Mean-rev (5d)	0.007	1.0	50.9
Volume (5d)	−0.046	−8.7	33.7
Low volatility	0.165	12.1	73.3
Random	0.001	0.4	53.3

5.4 Relationship to the Low-Volatility Anomaly

The low-volatility baseline achieves IC = 0.165, raising the question of whether Phantom simply rediscovers this known anomaly. We investigate via double-sorted portfolios: on each date, we independently sort assets into volatility terciles and Phantom prediction terciles, then examine returns within each cell (Figure 3a).

The cross-sectional correlation between Phantom's signal and the low-vol signal is $\rho = 0.63$, indicating substantial overlap. However, Phantom's ranking power *concentrates in the high-volatility tercile* (IC = 0.092, NW t = 6.77), precisely where sorting by volatility alone provides no differentiation—all high-vol assets earn negative relative returns, but Phantom distinguishes the "less bad" from the "worst." Combining both signals (equal-weighted rank average) yields the highest IC of 0.157, exceeding either signal alone.

5.5 Distributional Quality

Beyond ranking, the model produces calibrated uncertainty estimates (Figure 4). The PIT histogram is approximately uniform, and coverage calibration is excellent: 90% prediction intervals contain 90.3% of realized returns. The predicted volatility term structure follows $\sigma_h \propto \sqrt{h}$ at short horizons before diverging upward at longer horizons, reflecting the excess volatility clustering in crypto returns. The model consistently learns $\nu \approx 2$, the minimum degrees of freedom for finite variance, confirming that cryptocurrency relative returns have substantially heavier tails than a Gaussian model would accommodate.

5.6 Robustness

Figure 5 demonstrates the temporal stability of the signal. The rank IC is positive in all 15 out-of-sample months, ranging from 0.037

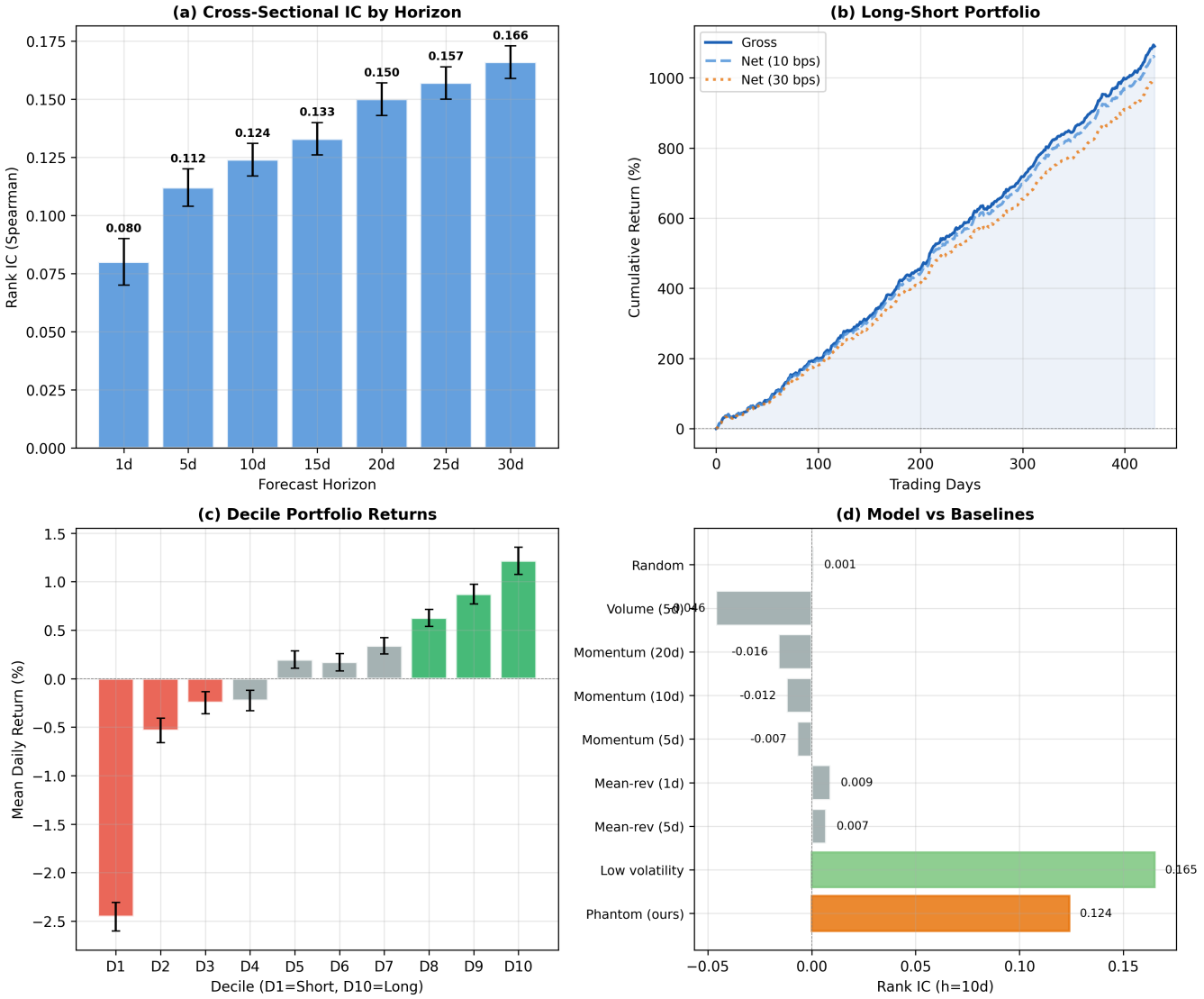

Figure 2: Main results. (a) Rank IC increases with horizon, consistent with drift SNR $\propto \sqrt{h}$. (b) Long-short cumulative returns under different cost assumptions. (c) Decile portfolio returns exhibit perfect monotonicity. (d) Phantom's IC dominates all simple baselines; the low-volatility strategy is the strongest competitor.

(March 2026, partial month) to 0.200 (November 2025). The 60-day rolling Sharpe ratio is positive 100% of the time, with a minimum of 2.82 and a mean of 14.0. Daily portfolio turnover is stable around 38%, with no regime-dependent spikes. The maximum drawdown is −38.6% with a 14-day recovery, occurring during a broad crypto deleveraging event.

5.7 Ablation: What Works and What Doesn't

Table 6 summarizes the progression across eight model versions, each testing a specific design hypothesis. The key inflection point is the shift from absolute to relative returns (v4→v5), which unlocks cross-sectional signal that was inaccessible in all prior versions. The most impactful positive finding is that widening the asset universe from 60 to 362 assets (v6→v8) doubles the Sharpe ratio through diversification, without any architectural change. Every negative result—synthetic pretraining, derivative features, and sub-daily data—eliminates a plausible design choice. Figure 6 shows the two-stage training dynamics for the final model.

6 Conclusion

We presented Phantom, a Transformer-based distributional forecasting model for cross-sectional cryptocurrency return prediction. By training on cross-sectionally demeaned (relative) returns across

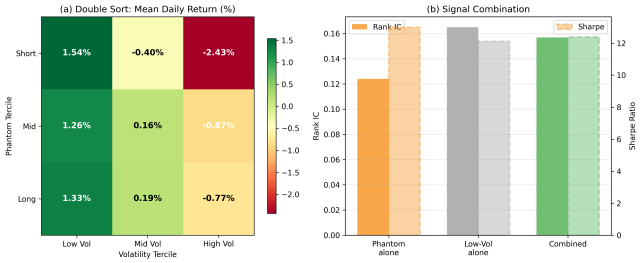

Figure 3: (a) Double sort: mean daily returns by volatility and Phantom tercile. The volatility column dominates (low-vol assets earn positive returns regardless of Phantom ranking), but Phantom's long-short spread is significant within mid-vol ($t = 2.6$) and high-vol ($t = 4.1$) terciles. (b) Combining both signals yields the highest IC.

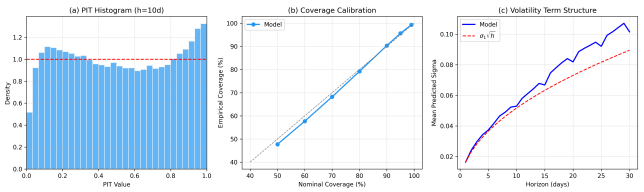

Figure 4: Distributional quality. (a) PIT histogram (h=10d) is approximately uniform. (b) Coverage calibration closely tracks the diagonal. (c) Predicted σ follows $\sqrt{h}$ scaling at short horizons.

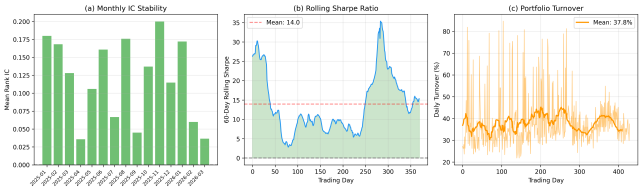

Figure 5: Robustness. (a) Monthly IC is positive in all 15 months. (b) 60-day rolling Sharpe always positive (min=2.82). (c) Turnover stable at 38%.

Table 6: Ablation across model versions (OOS crypto). "–" = not applicable.

Ver	Change	IC	Sharpe	Finding
v1	Synthetic pretrain	–	–	Oracle CRPS on synthetic
v2	+ Student-t head	–	–	Same CRPS, simpler
v3	Real multi-asset	0.00	–	Good calibration, no signal
v4	Multi-horizon abs.	0.00	–	Memorizes, doesn't transfer
v5	**Relative returns**	0.09	4.6	Signal unlocked
v6	+ Funding/taker buy	0.14	5.5	Features don't help
v7	4h bars	0.10	2.6	Worse: sample overlap
v8	**362 assets**	**0.12**	**13.0**	Sharpe doubles via breadth

362 cryptocurrencies, the model achieves a rank IC of 0.124 (Newey-West $t = 8.93$) with a long-short Sharpe ratio of 11.84 net of transaction costs, a 74.7% daily win rate, and perfect decile monotonicity on 15 months of fully out-of-sample data.

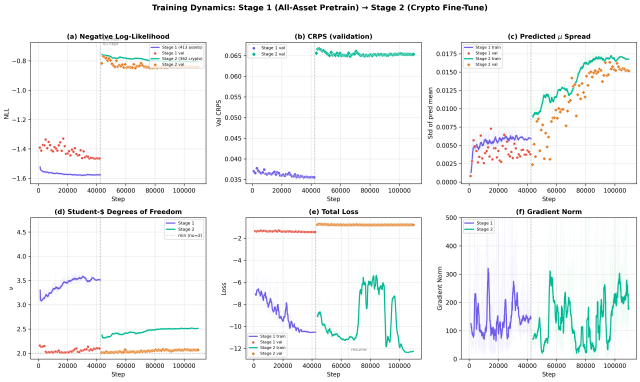

Figure 6: Two-stage training dynamics. Stage 1 (purple): pretraining on 413 multi-asset samples. Stage 2 (teal): fine-tuning on 362 crypto assets (dashed line = init from Stage 1 checkpoint). (a) NLL improves in both stages. (b) Val CRPS drops after crypto fine-tuning. (c) Predicted μ spread grows in Stage 2. (d) ν stabilizes near 2. (e) Total loss with resume point. (f) Gradient norm increases during adaptation.

The systematic ablation across eight model versions yields several insights: (i) absolute returns are unpredictable from OHLCV, but cross-sectional *relative* returns contain exploitable signal; (ii) synthetic SDE pretraining achieves oracle accuracy but transfers zero signal to real markets; (iii) derivative features add no value over price and volume; and (iv) signal scales with universe breadth but not with data frequency.

Limitations. The signal is crypto-specific—equities show near-zero IC—and our evaluation does not include live trading validation with real execution. The model's signal is partially correlated with the low-volatility anomaly ($\rho = 0.63$), and the distributional output (σ, ν) is not yet exploited for adaptive position sizing.

Future work. Promising directions include live deployment with execution-aware training, incorporation of alternative data (on-chain metrics, social sentiment), adaptive position sizing using the predicted $\sigma_{i,h}$, and extension to decentralized finance (DeFi) protocols where cross-sectional prediction over liquidity pools presents similar opportunities.

Reproducibility. All code, model checkpoints, and processed datasets are publicly available at https://github.com/ADnocap/phantom.

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A Model Hyperparameters

Table 7: Full model configuration for v8 (final model).

Hyperparameter	Value
Architecture	
Context length L	120 days
Input channels C	6 (OHLCV)
Patch length P	5
Number of patches N	24
Model dimension d	512
FFN dimension d_{ff}	2048
Encoder layers M	8
Decoder layers D	2
Attention heads	8
Forecast horizons H	30
Head hidden dimension	256
Total parameters	31.7M
Training	
Optimizer	AdamW (β_1=0.9, β_2=0.95)
Peak learning rate	3×10^{-4}
Min learning rate	1×10^{-6}
LR schedule	Cosine annealing
Warmup steps	2,000
Weight decay	0.05
Gradient clipping	1.0
Batch size	512
Dropout	0.3
Condition dropout p_{drop}	0.15
AMP	bfloat16
Loss weights	
λ_{NLL}	1.0
λ_{CRPS}	0.5
λ_{MSE}	0.3
λ_{var}	0.1
Min MSE horizon $h_{\min}$	5 days
Horizon weighting	$\sqrt{h/H}$
Output head	
σ range	$[e^{-7},\ e^{2}]$
ν range	$[2.01,\ 100]$